

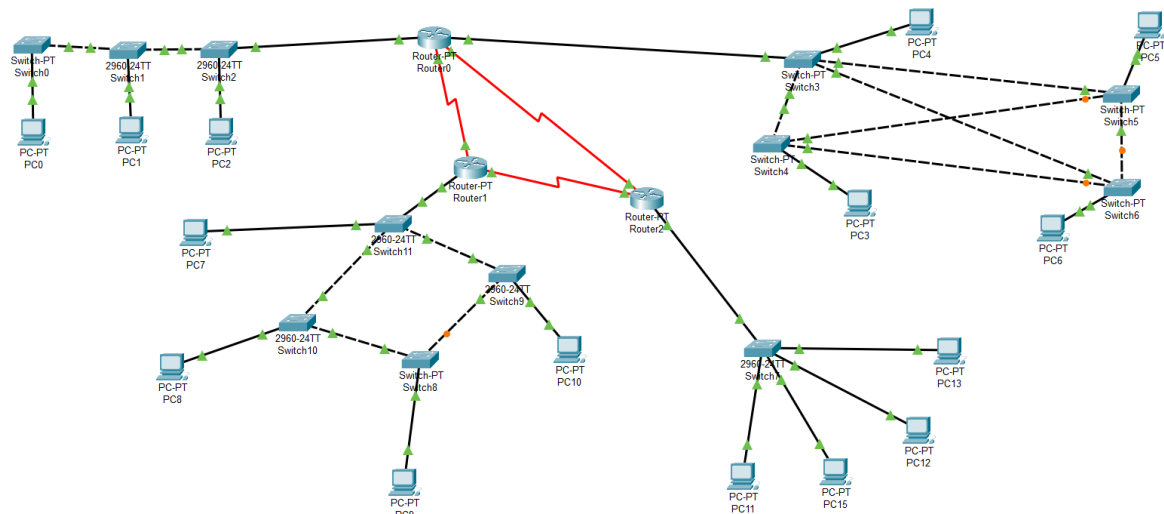
Computer Networks: Assignment-5

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Q 1) Construct the below network in the Cisco packet tracer, as shown below, and check where the data packets transfer from one system to another using

Diagram:



IP Config:

PC0: 192.168.1.4 / 255.255.255.0

PC1: 192.168.1.2 / 255.255.255.0

PC2: 192.168.1.3 / 255.255.255.0

PC3: 192.168.4.3 / 255.255.255.0

PC4: 192.168.4.5 / 255.255.255.0

PC5: 192.168.4.2 / 255.255.255.0

PC6: 192.168.4.4 / 255.255.255.0

PC7: 192.168.3.5 / 255.255.255.0

PC8: 192.168.3.2 / 255.255.255.0

PC9: 192.168.3.3 / 255.255.255.0

PC10: 192.168.3.4 / 255.255.255.0

PC11: 192.168.2.4 / 255.255.255.0

PC12: 192.168.2.2 / 255.255.255.0

PC13: 192.168.2.5 / 255.255.255.0

PC15: 192.168.2.3 / 255.255.255.0

Router0:

FastEthernet0/0: 192.168.1.1 / 255.255.255.0

FastEthernet1/0: 192.168.4.1 / 255.255.255.0

Serial2/0: 10.0.0.5 / 255.255.255.252

Serial3/0: 10.0.0.1 / 255.255.255.252

Router1:

FastEthernet0/0: 192.168.3.1 / 255.255.255.0

Serial2/0: 10.0.0.2 / 255.255.255.252

Serial3/0: 10.0.0.6 / 255.255.255.252

Router2:

FastEthernet0/0: 192.168.2.1 / 255.255.255.0

Serial2/0: 10.0.0.10 / 255.255.255.252

Serial3/0: 10.0.0.14 / 255.255.255.252

- i) Static routing (STATIC) and

Router Config:

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Network

Mask

Next Hop

Network Address

192.168.2.0/24 via 10.0.0.10

192.168.3.0/24 via 10.0.0.2

Router1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Network

Mask

Next Hop

Network Address

192.168.1.0/24 via 10.0.0.1

192.168.4.0/24 via 10.0.0.1

192.168.2.0/24 via 10.0.0.14

Router2

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Network

Mask

Next Hop

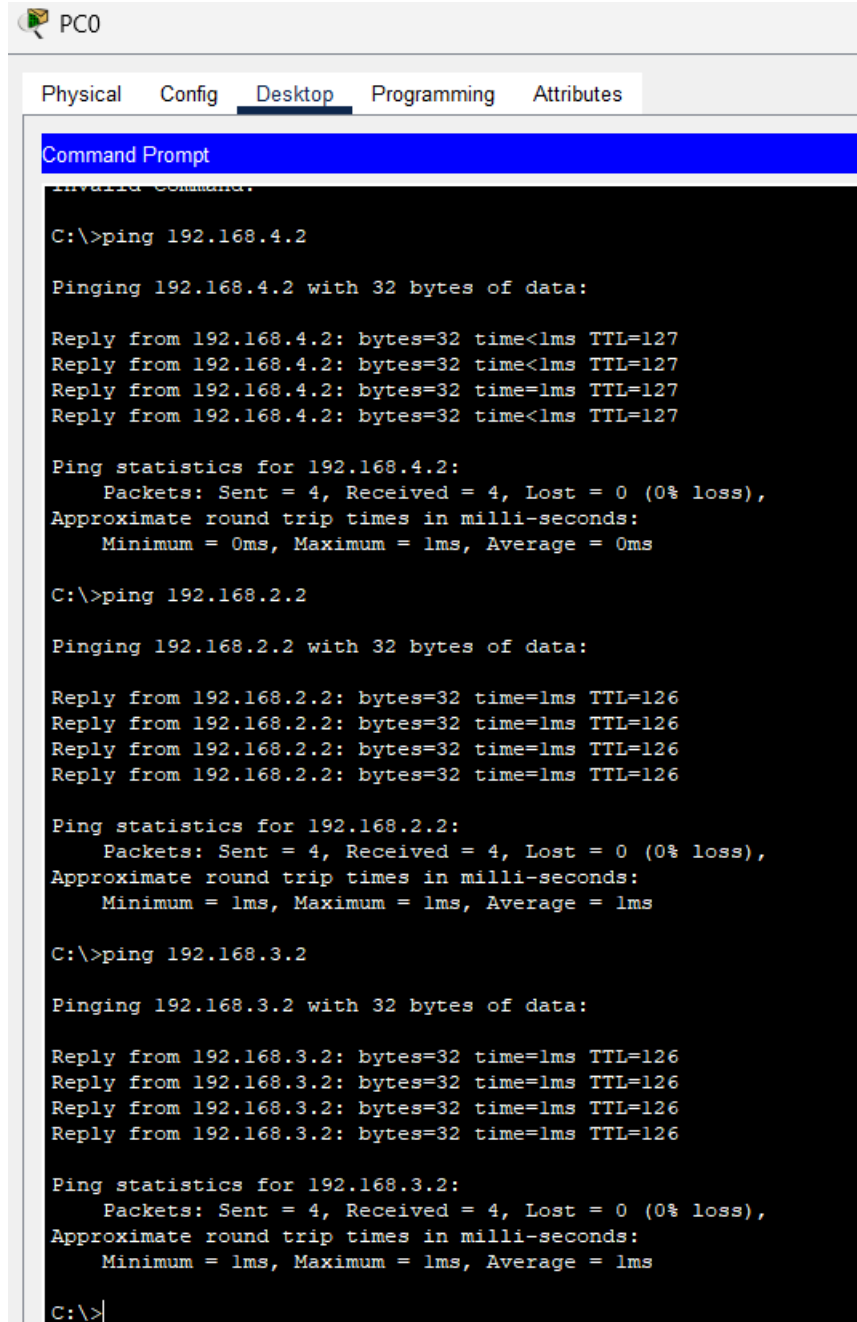
Network Address

192.168.4.0/24 via 10.0.0.5

192.168.1.0/24 via 10.0.0.5

192.168.3.0/24 via 10.0.0.6

Output:



The screenshot shows a PC0 desktop environment with a taskbar containing icons for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of three ping commands and their results.

```
Invalid Command.  
  
C:\>ping 192.168.4.2  
  
Pinging 192.168.4.2 with 32 bytes of data:  
  
Reply from 192.168.4.2: bytes=32 time<1ms TTL=127  
Reply from 192.168.4.2: bytes=32 time<1ms TTL=127  
Reply from 192.168.4.2: bytes=32 time=1ms TTL=127  
Reply from 192.168.4.2: bytes=32 time<1ms TTL=127  
  
Ping statistics for 192.168.4.2:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 1ms, Average = 0ms  
  
C:\>ping 192.168.2.2  
  
Pinging 192.168.2.2 with 32 bytes of data:  
  
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126  
  
Ping statistics for 192.168.2.2:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 1ms, Maximum = 1ms, Average = 1ms  
  
C:\>ping 192.168.3.2  
  
Pinging 192.168.3.2 with 32 bytes of data:  
  
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126  
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126  
  
Ping statistics for 192.168.3.2:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 1ms, Maximum = 1ms, Average = 1ms  
  
C:\>
```

- ii) Dynamic routing (RIP), further demonstrate the use of half and full duplex in the switch in the below network.

Router Config:

RIP Routing (v2)	
Network	
	Add
Network Address	
10.0.0.0	
192.168.1.0	
192.168.2.0	
192.168.3.0	
192.168.4.0	

Output:

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Invalid Command.

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time<1ms TTL=127
Reply from 192.168.4.2: bytes=32 time<1ms TTL=127
Reply from 192.168.4.2: bytes=32 time=1ms TTL=127
Reply from 192.168.4.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```